



SYSTEM DESIGN

Complete Learning Roadmap

Architect distributed systems with clear trade-off reasoning and operational reliability.

Scalability and reliability goals

Latency budgeting

Data consistency basics

API contracts

Caching strategies

Database trade-offs

01

Fundamentals

02

Core Concepts

03

Tools and Frameworks

04

Real Projects

05

Advanced Topics

5

Learning Stages

20+

Core Topics

10+

Projects

100%

System



Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Scalability and reliability goals
• Latency budgeting • Data consistency basics • API contracts

02

Core Concepts

Caching strategies • Database trade-offs • Message-driven systems • Failure handling patterns

03

Tools and Frameworks

Queue technologies • Observability stack • Load balancing systems • Service gateways

04

Real Projects

URL shortener design • Notification platform design • Feed service design • Realtime chat design

05

Advanced Topics

Global multi-region architecture • Capacity planning • Chaos testing • Cost/performance trade-off governance

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Scalability and reliability goals
- Latency budgeting
- Data consistency basics
- API contracts

Tools & Frameworks

- Scalability and reliability goals
- Latency budgeting
- Data consistency basics
- API contracts

Recommended Projects

- System Design workshop using Scalability and reliability goals
- System Design portfolio project with Latency budgeting

Stage 1 Milestone Complete System Design workshop using Scalability and reliability goals and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Caching strategies
- Database trade-offs
- Message-driven systems
- Failure handling patterns

Tools & Frameworks

- Caching strategies
- Database trade-offs
- Message-driven systems
- Failure handling patterns

Recommended Projects

- System Design workshop using Caching strategies
- System Design portfolio project with Database trade-offs

Stage 2 Milestone Complete System Design workshop using Caching strategies and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- Queue technologies
- Observability stack
- Load balancing systems
- Service gateways

Tools & Frameworks

- Queue technologies
- Observability stack
- Load balancing systems
- Service gateways

Recommended Projects

- System Design workshop using Queue technologies
- System Design portfolio project with Observability stack

Stage 3 Milestone Complete System Design workshop using Queue technologies and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- URL shortener design
- Notification platform design
- Feed service design
- Realtime chat design

Tools & Frameworks

- URL shortener design
- Notification platform design
- Feed service design
- Realtime chat design

Recommended Projects

- System Design workshop using URL shortener design
- System Design portfolio project with Notification platform design

Stage 4 Milestone Complete System Design workshop using URL shortener design and document the decisions, trade-offs, and validation checks before moving to Stage 5.



Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

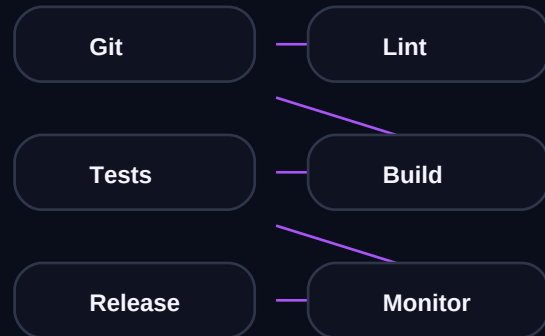
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Global multi-region architecture
- Capacity planning
- Chaos testing
- Cost/performance trade-off governance

CI/CD Pipeline



Stage 5 Milestone

A production-grade system design workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

System Design docs and community guides

Link: <https://martinfowler.com/>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/system-design>

PAID

Structured specialization track focused on System Design

Link: <https://roadmap.sh/system-design>

FREE

System Design roadmap reference

Link: <https://roadmap.sh/system-design>

FREE

Official documentation

Link: <https://martinfowler.com/>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in System Design?
- Which project best proves Stage 4 competency in System Design?
- What trade-offs appear in Stage 5 decisions for System Design?